

INTRODUCTION TO DATA ANALYSIS WITH PYTHON

Session 1: Kickoff & Python as a calculator

Course overview, grading, and doing the first steps with Python

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About us



Silvia Tulli

Associate Professor, Sorbonne University

Head of AI Research, HABS

Research focus: AI and robotics, multimodal machine learning.

Teaches sessions on importing and cleaning data, data manipulation, and a recap session.



Cecilia Graiff

Ph.D. Student, ALMANaCH, Inria Paris

Research focus: AI, NLP, machine learning.

Teaches sessions on introduction to Python, functions, data structures, NumPy and Pandas, data manipulation, exploratory data analysis



Paul Jeu

Ph.D. Student, Université Paris Cité

Research focus: Ethics of AI

Teaches sessions on control flow, lists, and the first recap session.

Today's agenda

- 01 Welcome & introductions**
Who we are, how the course works
- 02 Course overview**
Description, prerequisites, expected skills, format
- 03 Grading**
How the course is evaluated, and what the final project involves
- 04 Detailed syllabus**
All 12 sessions, topic by topic
- 05 Election of class representative**
Your point of contact with us throughout the semester
- 06 Technical session: Jupyter Notebook & Python as a calculator**
Coding tools, basic Python, numbers, strings, lists

Wooclap Kickoff

- Event Code: JGDFPNJ



Introduction to Data Analysis with Python

Building on core Python skills, this course introduces data analysis and visualization using pandas

- Work with real-world datasets provided in clean, ready-to-use form
- **Learn the fundamentals of loading, filtering, and summarizing data with pandas**
- Build clear, effective visualizations to explore trends and patterns
- Practice writing simple, readable code to answer real questions about data

Collect

Clean

Explore

Visualize

Summarize

Prerequisites and expected skills

COMING IN, YOU SHOULD HAVE

- Familiarity with basic statistics — concepts like mean, median, and standard deviation are helpful but not mandatory
- An interest in working with data
- Willingness to practice regularly

BY THE END, YOU'LL BE ABLE TO

- Python core concepts: data types, control structures, functions, error handling
- Data handling with pandas: opening, cleaning, and manipulating datasets; basic numpy
- Basic statistics: summary statistics, simple linear regression
- Basic visualization: create maps, represent data in form of diagrams

Wooclap Prerequisites

- Event Code: MGQYYOF



Academic format and materials

- **Materials are shared on Moodle and GitHub** — slides and Jupyter notebooks
- Each session opens with a short theoretical part, then moves into hands-on coding in notebooks
- **Expect a mix of** individual exercises, small-group work, and live Wooclap quizzes

<https://github.com/cg67qowo/Advanced-Python>
<https://moodle.sciences-po.fr/course/view.php?id=64748>

2h

in class, every week

12

sessions this semester

~2h50

expected work per week outside class

How the course is graded



20% — Continuous assessment. Short in-class quizzes (Wooclap) throughout the semester, checking understanding session by session.

80% — Final project. Submitted at the end of the course; evaluated on both the practical output and a written report.

Validation mode: the final project

- **Content** — a data analysis project applying the methods covered in the main course (regression, classification, or time series), on a topic agreed together with the professors during the semester
- **Due** — submitted at the end of the course (exact date confirmed via Moodle)
- **Criteria** — practical output (code, models, data visualizations) and a final written report
- **Format** — details on individual vs. group work to be confirmed as the topic is set

WHAT TO KEEP IN MIND

The project topic connects directly to what you're studying in your main policy/analysis courses — **start thinking early about a dataset or question you care about.**

Detailed syllabus — 12 sessions

01	Introduction to Python	Cecilia Graiff
02	Control Flow	Paul Jeu
03	Defining Functions	Cecilia Graiff
04	Data Structures: Lists	Paul Jeu
05	Data Structures: Tuples, Sets, and Dictionaries	Cecilia Graiff
06	Recap: Types, Functions, and Structures	Paul Jeu
07	NumPy and Pandas	Cecilia Graiff
08	Importing and Cleaning Data	Silvia Tulli
09	Data Manipulation	Cecilia Graiff
10	Data Visualization	Silvia Tulli
11	Exploratory Data Analysis	Cecilia Graiff
12	Recap	Silvia Tulli

Why we're electing a class representative

- Acts as the main point of contact between the class and us throughout the semester
- We'll turn to them for quick, informal checks — e.g. a poll on pacing, or scheduling group work
- **They gather and relay class feedback at the mid-semester point**, so we can adjust before it's too late
- It's a light commitment: mostly relaying messages both ways

CLASS REPRESENTATIVE

Let's elect one now

GETTING STARTED

Google Colab

Python, in your browser — nothing to install

1

Zero Setup

Runs entirely in the cloud. Open a browser tab and you're ready to code.

2

Notebook Format

Mixes code, text, and output together, so your work reads like a report.

3

Ready to Go

Python and common libraries like pandas come preloaded — no install.

During our classes, we will be using Colab notebooks, where you will be able to code directly. Let's switch to our first one now!

WRAP-UP

Before next session

- Open a Colab session on your laptop and play around with it
- Do the exercises on the notebook
- Re-read the notebook and repeat the core concepts
- Reach out to your class representative with any early questions